

Original relation moments and their positive-form comparison

Split-Zero research programme

13 September 2026

This supplement gives the complete finite proof behind the incoming relation-moment certificate, records its two domain corrections, and constructs the exact source factorization for an upper comparison kernel. Every quotient coordinate, original source norm, zero eigenvalue, and source label is retained. The comparison kernel is additional data on the existing coefficient space; it does not replace the arithmetic norm. The arithmetic construction of that kernel is supplied by the cumulative paper's arithmetic upper-comparison proof.

1 The original Hilbert objects and their maps

Fix a monic polynomial $\chi \in \mathbb{C}[S]$ of degree $q \geq 1$ and retain $E = \mathbb{C}[S]/(\chi)$ with ordered remainder basis $1, S, \dots, S^{q-1}$. The original multiplication endomorphism is $A = M_S$. Fix the original monic orthogonal polynomial columns p_n , their literal positive squared norms ω_n , and $b_n = [p_n]_\chi$. The degree- N source $\mathcal{P}_N = (\mathbb{C}^{N+1}, O_N)$ has $O_N = \text{diag}(\omega_0, \dots, \omega_N)$ and quotient map $B_N = [b_0, \dots, b_N]$. For $N \geq q - 1$, its first q columns form an invertible triangular matrix, so B_N is surjective and

$$K_N = B_N O_N^{-1} B_N^*, \quad G_N = K_N^{-1}, \quad R_N = O_N^{-1} B_N^* G_N, \quad V_N = \det G_N > 0. \quad (\text{RM.1})$$

Here K_N is a positive form on the conjugate dual coefficient space and G_N is the original quotient inner product, conjugate-linear in its first argument. Direct multiplication proves $B_N R_N = 1$ and $R_N^* O_N R_N = G_N$. If $u \in \ker B_N$, then $u^* O_N R_N x = u^* B_N^* G_N x = 0$. Thus R_N is the actual least-norm lift.

Fix $q - 1 \leq i \leq j$, set $r = j - i$, and write

$$F = [b_{i+1}, \dots, b_j], \quad \Omega = \text{diag}(\omega_{i+1}, \dots, \omega_j), \quad D = K_j - K_i = F \Omega^{-1} F^*. \quad (\text{RM.2})$$

For $r = 0$ these maps have their literal zero-dimensional domains. The identity $I : E_i = (E, G_i) \rightarrow E_j = (E, G_j)$ has adjoint $T = K_i G_j : E_j \rightarrow E_i$, since $G_i T = G_j$. Consequently

$$Q = T I \in \text{End}(E_i), \quad Z = Q^{-1} - 1 = D G_i = F F^\dagger, \quad F^\dagger = \Omega^{-1} F^* G_i. \quad (\text{RM.3})$$

In particular $G_i Z = G_i D G_i \succeq 0$, so Z is positive self-adjoint in the original G_i . With its full q eigenvalues $\lambda_a \geq 0$,

$$g_a = (1 + \lambda_a)^{-1}, \quad \frac{V_i}{V_j} = \det(1 + Z) = \prod_{a=1}^q (1 + \lambda_a). \quad (\text{RM.4})$$

All zero λ_a contribute the factor one and remain in the dimension. If $Z = 0$, then $V_i = V_j > 0$, the ratio equals one, and its logarithm equals zero. This corrects the incoming note's phrase "the volume is zero."

For the real scalar k in the arithmetic action identity, put $H_N = K_N A^* G_N + A - k1$. Regard H_j on E_i via the specified coordinate identity and denote that endomorphism by $\hat{H}_j$. Calculation in the original coordinates gives

$$\begin{aligned} H_i Q - Q \hat{H}_j &= (K_i A^* G_i + A - k1) K_i G_j - K_i G_j (K_j A^* G_j + A - k1) \\ &= A K_i G_j - K_i G_j A = [A, Q]. \end{aligned}$$

Multiplying $[A, Q^{-1}] = -Q^{-1}[A, Q]Q^{-1}$ out proves

$$[A, Z] = \hat{H}_j(1 + Z) - (1 + Z)H_i. \quad (\text{RM.5})$$

Thus the cost and action are related even when their eigenspaces differ.

2 The source correction, all cross-pairings, and exterior area

The map $\Phi = R_i F : (\mathbb{C}^r, \Omega) \rightarrow \mathcal{P}_i$ has image Gram $\Phi^* O_i \Phi = F^* G_i F$. Trace cyclicity and Hermitian symmetry give

$$t_1 = \text{Tr } Z = \sum_{n=i+1}^j \frac{b_n^* G_i b_n}{\omega_n}, \quad (\text{RM.6})$$

$$t_2 = \text{Tr } Z^2 = \sum_{n,m=i+1}^j \frac{|b_n^* G_i b_m|^2}{\omega_n \omega_m}, \quad (\text{RM.7})$$

$$\mathcal{A}_2 = \frac{t_1^2 - t_2}{2} = \sum_{i < n < m \leq j} \frac{(b_n^* G_i b_n)(b_m^* G_i b_m) - |b_n^* G_i b_m|^2}{\omega_n \omega_m}. \quad (\text{RM.8})$$

The diagonal terms cancel in $t_1^2 - t_2$; each unordered pair occurs twice, proving the last identity with its literal factor $1/2$. In unscaled exterior bases endowed with determinant Gram matrices,

$$\|\Phi\|_{\text{HS}}^2 = t_1, \quad \left\| \bigwedge^2 \Phi \right\|_{\text{HS}}^2 = \mathcal{A}_2. \quad (\text{RM.9})$$

Indeed the source vector $e_n \wedge e_m$ has squared norm $\omega_n \omega_m$ and its image has exactly the numerator in (RM.8). The alternating tensor map $u \wedge v \mapsto u \otimes v - v \otimes u$ has image Gram twice the exterior Gram. Applying this map in both source and target yields the commutative exterior/tensor square and cancels that same factor in the Hilbert–Schmidt ratio; neither Gram is identified with the other without the factor.

Let $L : \mathcal{P}_i \rightarrow \mathcal{P}_j$ pad by zero and let $N : (\mathbb{C}^r, \Omega) \rightarrow \mathcal{P}_j$ be the actual new-column inclusion. Their images are orthogonal and $B_j N = F$, $B_j L = B_i$. Hence the retained relation map

$$\mathbf{b} = N - L R_i F, \quad B_j \mathbf{b} = 0, \quad \mathbf{b}^* O_j \mathbf{b} = \Omega + F^* G_i F. \quad (\text{RM.10})$$

This proves the relation identity and its full norm. Under the original source realization $\mathcal{V}$, the equality $J^{(k)} \mathcal{V} = \eta \pi_\chi$ makes both components have the same arithmetic jets. The original internal quotient sends $(\ell_j, \mathbf{b}v)$ to $(\ell_j, 0)$, preserving the receiving label. Its lift sends external absence τ to τ . The coefficient amplitude map $p_R : G(R) \rightarrow R$ sends both $0_R^\bullet$ and τ to 0_R , and the exact fibre $p_R^{-1}(0_R) = \{0_R^\bullet, \tau\}$ records precisely the information lost by this subsequent observation. Likewise a dependent admitted wedge remains an admitted source vector whose observed squared area is $0^\bullet$.

3 Sharp determinant bound, exact remainder, and all boundary cases

For $q \geq 2$, positivity gives $t_1^2/q \leq t_2 \leq t_1^2$ by Cauchy–Schwarz and the nonnegative pair products. Define

$$d = \sqrt{\frac{qt_2 - t_1^2}{q-1}}, \quad a = \frac{t_1 + (q-1)d}{q}, \quad b = \frac{t_1 - d}{q}. \quad (\text{RM.11})$$

Thus $0 \leq d \leq t_1$, $0 \leq b \leq a$, and direct expansion proves $a + (q-1)b = t_1$ and $a^2 + (q-1)b^2 = t_2$. For any eigenvalue x , the other $q-1$ eigenvalues give $t_2 \geq x^2 + (t_1 - x)^2/(q-1)$. Solving that quadratic yields $x \leq a$. Consequently $Q \succeq (1+a)^{-1}1$ in G_i .

Here is a direct algebraic proof of the entire interpolation remainder. For $s \geq 1$, let

$$H_s(x) = \frac{1}{s+b} - \frac{x-b}{(s+b)^2} + \frac{(x-b)^2}{(s+a)(s+b)^2}. \quad (\text{RM.12})$$

Its value and derivative agree with $1/(s+x)$ at b , and its value agrees at a . Multiplication by the common denominator proves, identically,

$$\frac{1}{s+x} - H_s(x) = \frac{(a-x)(x-b)^2}{(s+x)(s+b)^2(s+a)}. \quad (\text{RM.13})$$

Integrate $1/s - 1/(s+x)$ over $[1, \infty)$ to obtain $\log(1+x)$. Integration of $1/s - H_s(x)$ gives its quadratic interpolant

$$P(x) = \log(1+b) + \frac{x-b}{1+b} + c(x-b)^2, \\ c = \frac{\log(1+a) - \log(1+b) - (a-b)/(1+b)}{(a-b)^2} < 0 \quad (a > b). \quad (\text{RM.14})$$

The coefficient integrals converge: the quadratic coefficient is $-\int_1^\infty ((s+a)(s+b)^2)^{-1} ds$, and the other two are the displayed logarithm and reciprocal. Therefore, for $0 \leq x \leq a$,

$$P(x) - \log(1+x) = (a-x)(x-b)^2 \int_1^\infty \frac{ds}{(s+x)(s+b)^2(s+a)} \geq 0. \quad (\text{RM.15})$$

The trace of any quadratic depends only on q, t_1, t_2 . Those values equal the moments of $a, b, \dots, b$, so summing (RM.15) proves

$$\log(V_i/V_j) \leq U_2 := \log(1+a) + (q-1)\log(1+b). \quad (\text{RM.16})$$

The remainder in (RM.15) is strictly positive away from $x = a, b$. If m eigenvalues equal a , the first trace identity gives $ma + (q-m)b = a + (q-1)b$, hence $(m-1)(a-b) = 0$. Equality for $a > b$ therefore occurs exactly once at a and $q-1$ times at b . That spectrum realizes the prescribed moments, proving sharpness from these data. For $a = b$, zero variance forces $Z = b1$ and equality already holds. For $q = 1$, the ratio is exactly $1 + t_1$. For $q = 0$, its empty determinant is one. The case $i = j$ gives $Z = 0$ at every admitted dimension.

With $t_3 = \text{Tr } Z^3$, set

$$C_3 = -t_3 + (a+2b)t_2 - (2ab+b^2)t_1 + qab^2 = \text{Tr}((a1-Z)(Z-b1)^2) \geq 0. \quad (\text{RM.17})$$

Finite spectral summation of (RM.15) proves the exact loss

$$U_2 - \log \det(1+Z) = \int_1^\infty \frac{\text{Tr}((a1-Z)(Z-b1)^2(s1+Z)^{-1})}{(s+b)^2(s+a)} ds. \quad (\text{RM.18})$$

Every factor is a function of the same G_i -self-adjoint endomorphism. Since $(s+x)(s+b)^2(s+a) \leq (s+a)^4$, the loss is at least $C_3 \int_1^\infty (s+a)^{-4} ds = C_3/(3(1+a)^3)$. Also Jensen's scalar inequality, whose strictness follows from $\frac{d^2}{dx^2} \log(1+x) = -(1+x)^{-2} < 0$, gives $U_2 \leq q \log(1+t_1/q)$, strictly unless $a=b$.

For $q \geq 2$ and $t_1 > 0$, $\mathcal{A}_2 = (q-1)b(t_1 - qb/2)$ and $0 \leq b \leq t_1/q$ imply

$$\frac{\mathcal{A}_2}{(q-1)t_1} \leq b \leq \frac{2\mathcal{A}_2}{(q-1)t_1}, \quad \frac{V_i}{V_j} \leq (1+t_1) \left(1 + \frac{2\mathcal{A}_2}{(q-1)t_1}\right)^{q-1}. \quad (\text{RM.19})$$

When $\mathcal{A}_2 = 0 < t_1$, all eigenvalue pair products vanish, so exactly one eigenvalue is positive and the ratio is $1+t_1$.

4 The comparator join in the original metric

Let U_j be the actual explicit upper kernel supplied by the arithmetic comparison construction, expressed in the same quotient coordinates. That construction proves $U_j \succeq K_j \succeq K_i$ as dual forms. Define from the available lower-degree arithmetic data and this comparator

$$D_+ = U_j - K_i, \quad Z_+ = D_+ G_i, \quad t_1^+ = \text{Tr } Z_+, \quad t_2^+ = \text{Tr } Z_+^2. \quad (\text{RM.20})$$

Then $G_i(Z_+ - Z) = G_i(U_j - K_j)G_i \succeq 0$. Thus both operators are positive self-adjoint in G_i and $Z_+ \succeq Z$ there. For completeness, ordered eigenvalues increase under this order: for an orthonormal eigenbasis of either operator, the top- m span gives the lower inequality below; every m -dimensional subspace intersects the span of eigenvectors numbered $m, \dots, q$, which gives the upper inequality. This proves the max-min characterization

$$\lambda_m(X) = \max_{\dim M=m} \min_{v \in M \setminus 0} \frac{\langle v, Xv \rangle_{G_i}}{\langle v, v \rangle_{G_i}}.$$

Applying the pointwise form inequality inside this characterization yields $\lambda_m(Z_+) \geq \lambda_m(Z)$ for each of the original q indices. In particular, for $q \geq 2$, with a_+, b_+ obtained by (RM.11) from t_1^+, t_2^+ ,

$$\frac{V_i}{V_j} \leq \det(1 + Z_+) = \det(U_j G_i) \leq (1+a_+)(1+b_+)^{q-1}. \quad (\text{RM.21})$$

No monotonicity of the sharp bound as a function of two separately varied trace numbers was used; the first inequality follows from the actual form order, and the second follows from (RM.16) for Z_+ . For $q \geq 2$, $Q \succeq (1+a_+)^{-1}1$ is a derived finite gap. For $q=1$, the comparator determinant is $1+t_1^+$ and $Q \succeq (1+t_1^+)^{-1}1$; for $q=0$, the empty determinant is one. When $i \leq q$, the actual input to K_i, G_i uses moments only through $2i \leq 2q$; U_j uses the explicit comparison measure's moments through $2j \leq 4q$ in the two endpoint blocks. Formula (RM.21) supplies a certificate without introducing the unknown actual high-degree moments into its inputs.

There is an exact source map behind this comparison. Interpret D_+ as a positive Hermitian form on the conjugate dual coefficient space $E^\vee$. Let

$$\mathcal{C}_+ = E^\vee / \ker D_+, \quad \langle [\alpha], [\beta] \rangle_+ = \alpha^* D_+ \beta, \quad J_+[\alpha] = D_+ \alpha \in E_i. \quad (\text{RM.22})$$

These expressions are independent of representatives. Positivity implies the only zero-norm classes are the removed radical: diagonalize the positive Hermitian form in the stated coordinate space to

prove $\alpha^* D_+ \alpha = 0 \iff D_+ \alpha = 0$. This diagonalization proves a fact about that form and does not alter G_i . The adjoint is $J_+^\dagger v = [G_i v]$, hence $J_+ J_+^\dagger = Z_+$ and

$$\Phi_+ = R_i J_+ : \mathcal{C}_+ \rightarrow \mathcal{P}_i, \quad \|\Phi_+\|_{\text{HS}}^2 = t_1^+, \quad \|\bigwedge^2 \Phi_+\|_{\text{HS}}^2 = ((t_1^+)^2 - t_2^+)/2. \quad (\text{RM.23})$$

To prove the second identity in any Gram basis, diagonalize the positive endomorphism $\Phi_+^\dagger \Phi_+$ on its existing source Hilbert space. Its exterior eigenvalues are all unordered products. Its nonzero eigenvalues agree, with multiplicity, with those of $\Phi_+ \Phi_+^\dagger$ by the mutually inverse maps Φ_+ and $\lambda^{-1} \Phi_+^\dagger$ on each nonzero eigenspace. Finally R_i is an isometry from E_i into $\mathcal{P}_i$, which proves (RM.23), retaining the zero eigenspaces.

The original correction factors through this source by an exact contraction. For $\alpha \in \ker D_+$, (RM.2) and $D \preceq D_+$ imply $\alpha^* F \Omega^{-1} F^* \alpha = 0$, so $F^* \alpha = 0$. Thus $Fv \in \text{im } D_+$ for every v . There is a unique class $Cv = [u]$ solving $D_+ u = Fv$. This defines a linear map

$$C : (\mathbb{C}^r, \Omega) \rightarrow \mathcal{C}_+, \quad J_+ C = F, \quad \Phi_+ C = \Phi. \quad (\text{RM.24})$$

Let $s = u^* D_+ u \geq 0$. The equality $D_+ u = Fv$ and Cauchy–Schwarz in Ω give $s = |u^* Fv| \leq (u^* F \Omega^{-1} F^* u)^{1/2} (v^* \Omega v)^{1/2} \leq s^{1/2} (v^* \Omega v)^{1/2}$. Dividing when $s > 0$, with the case $s = 0$ immediate, proves $\|Cv\|_+^2 \leq v^* \Omega v$. The full kernel is $\ker C = \ker F$ by (RM.24) and uniqueness of the class. Exterior functoriality gives $\bigwedge^2 \Phi = (\bigwedge^2 \Phi_+) (\bigwedge^2 C)$, preserving every admitted pair. This is the exact morphism from the old source correction to the new positive-form source. The original relation $N - L\Phi$ from (RM.10) is retained, with its original supported-zero image.

The cost-action defect of the comparator also has an explicit form. Put $\Delta = U_j - K_j$. Then $Z_+ = Z + \Delta G_i$, and direct expansion using $\hat{H}_j = A + K_j A^* G_j - k1$ gives

$$\begin{aligned} [A, Z_+] &= \hat{H}_j (1 + Z_+) - (1 + Z_+) H_i + \mathcal{E}, \\ \mathcal{E} &= [A, \Delta G_i] - \hat{H}_j \Delta G_i + \Delta G_i H_i \\ &= \Delta A^* G_i - K_j A^* G_j \Delta G_i. \end{aligned} \quad (\text{RM.25})$$

The second line defines the exact defect; substituting the displayed control matrices and cancelling equal terms proves the last expression. It is not discarded or asserted small.

5 Literal Gaussian calibration and source corrections

For the declared polynomial calibration, retain $S = 1 + ix$, literal Gaussian mass seven, variance one, $p_{n+1} = (S - 1)p_n + np_{n-1}$ and $\omega_n = 7n!$. For $q = 3$, write $y = S - 1$ only as the explicit polynomial $S - 1$, without changing the remainder basis. The relevant columns, expressed in the fixed basis $1, S, S^2$, are

$$\begin{aligned} b_0 &= (1, 0, 0)^t, \quad b_1 = (-1, 1, 0)^t, \quad b_2 = (2, -2, 1)^t, \quad b_3 = (-3, 3, 0)^t, \\ b_4 &= (9, -12, 6)^t, \quad b_5 = (-15, 15, 0)^t, \quad b_6 = (60, -90, 45)^t. \end{aligned}$$

These follow by the literal recurrence and division by $(S - 1)^3$. Substitution into (RM.1)–(RM.3) gives

$$Z_{3,6} = \frac{1}{8} \begin{pmatrix} -29 & -35 & 84 \\ 42 & 48 & -126 \\ -21 & -21 & 69 \end{pmatrix}. \quad (\text{RM.26})$$

The direct recurrence gives $G_3 = \begin{pmatrix} 7 & 7 & 0 \\ 7 & 49/5 & 28/5 \\ 0 & 28/5 & 126/5 \end{pmatrix}$, so (RM.26) is its literal coefficient matrix in the retained metric. Its invariant conclusions are

$$t_1 = 11, \quad t_2 = 3365/32, \quad t_3 = 273947/256, \quad \mathcal{A}_2 = 507/64, \quad \det(1 + Z) = 5145/256. \quad (\text{RM.27})$$

The characteristic polynomial factors with roots $3/4, (41 - 7\sqrt{34})/8, (41 + 7\sqrt{34})/8$; all are positive, distinct, and retain $q = 3$. The two-trace extremal roots are $a = 11/3 + 7\sqrt{127}/12$, $b = 11/3 - 7\sqrt{127}/24$. Since the actual three roots are distinct and the equality spectrum in (RM.16) has two equal roots, its inequality is strict. Since $a > b$, its improvement over $3\log(1 + 11/3)$ is also strict. Rational logarithm enclosures in the companion replay give the numerical values in the incoming note without relying on decimals for either strictness argument.

In dimension two, the same literal recurrence gives $(t_1, t_2) = (2, 5/2)$ for $(i, j) = (1, 3)$ and $(7/4, 37/16)$ for $(2, 4)$. The respective (a, b) are $(3/2, 1/2)$ and $(3/2, 1/4)$, so (RM.16) is exact and yields ratios $15/4$ and $25/8$.

The incoming window expressions contain $1/(2r)$ and a minimum over $i \leq N < j$. Their domain is $i < j$. The original and comparator maps retain $i = j$ as well; scalar formulas keep their explicitly stated dimension and positive-trace domains, with zero cost handled separately. No minimum over an empty window is defined, and no inverse below the first admissible degree $q - 1$ is introduced. The complete first-degree and consecutive-window proofs remain in the cumulative reader.

6 Rank information with every zero direction retained

Let $W = \text{im } F \subset E_i$ and $\rho = \dim W$. Since $\langle v, Zv \rangle_{G_i} = \|F^\dagger v\|_\Omega^2$, one has

$$\ker Z = \ker F^\dagger = W^{\perp_{G_i}}, \quad E_i = W \oplus^{\perp_{G_i}} W^{\perp_{G_i}}. \quad (\text{RM.28})$$

To check invariance and positivity on W , $Z = FF^\dagger$ has image in W , and the only vector of W in its kernel is zero by (RM.28). The restriction Z_W is therefore positive definite on its inherited metric, and the literal inclusion/projection maps give

$$Z = Z_W \oplus 0_{q-\rho}, \quad Q = (1_W + Z_W)^{-1} \oplus 1_{q-\rho}, \quad \det(1_E + Z) = \det(1_W + Z_W) 1^{q-\rho}. \quad (\text{RM.29})$$

The source kernel is also retained. Put $\mathcal{U} = (\ker F)^\perp_\Omega$ in the original source. Its orthogonal projection $P_\mathcal{U}$ satisfies $F = F|_\mathcal{U} P_\mathcal{U}$. The map $F|_\mathcal{U} : \mathcal{U} \rightarrow W$ is a bijection: injectivity is the orthogonal intersection with $\ker F$, and surjectivity follows by decomposing any source preimage. Thus it identifies the source Hilbert quotient $(\mathbb{C}^r, \Omega)/\ker F$ with W by a specified map; the norms on these objects are related by that map's retained positive Gram, rather than being asserted equal. Its composition with R_i is precisely the old least-lift correction on the quotient source. For $v \in \ker F$, the original relation (RM.10) is Nv , so the kernel directions remain actual newly admitted theta relations, with their norm $v^* \Omega v$ and their receiving supported-zero labels.

Any proved integer upper bound $\rho \leq R \leq q$ gives a sharper finite determinant certificate. The ρ positive eigenvalues and $R - \rho$ expressly retained zeros have the same t_1, t_2 . For $R \geq 2$ set

$$d_R = \sqrt{\frac{Rt_2 - t_1^2}{R - 1}}, \quad a_R = \frac{t_1 + (R - 1)d_R}{R}, \quad b_R = \frac{t_1 - d_R}{R}. \quad (\text{RM.30})$$

Applying the already proved scalar argument to that R -entry list gives

$$\det(1_E + Z) \leq (1 + a_R)(1 + b_R)^{R-1} 1^{q-R}. \quad (\text{RM.31})$$

This operation does not identify or remove the $q - R$ original directions. The full decomposition (RM.29) records their action by Q and Z . For $R = 1$, the exact ratio is $1 + t_1$; for $R = 0$, it is one. One always has $\rho \leq \min(q, j - i)$ from the literal dimensions of F . Additional parity or relation identities may give a better proved bound; their full proofs must supply that information. The present argument makes no such unproved rank assertion for an arithmetic packet.

For fixed feasible t_1, t_2 , this bound is at least as sharp as the q -dimensional bound. Indeed the R -entry extremizer $a_R, b_R, \dots, b_R$ with $q - R$ additional zeros is feasible for the q -dimensional moment problem. The optimality already proved for that problem yields

$$\log(1 + a_R) + (R - 1) \log(1 + b_R) \leq \log(1 + a) + (q - 1) \log(1 + b).$$

The known rank bound belongs to the actual cost Z . To apply (RM.30)–(RM.31) to Z_+ , its own rank must be proved; the inequality $Z_+ \succeq Z$ does not imply $\text{rank } Z_+ \leq \text{rank } Z$. Its exact source in (RM.22) gives $\text{rank } Z_+ = \dim \mathcal{C}_+ = \text{rank } D_+$, which is the correct replacement.

7 The entire comparison excess is the source contraction defect

The contraction in (RM.24) has a concrete adjoint:

$$C^\dagger[\alpha] = \Omega^{-1} F^* \alpha. \quad (\text{RM.32})$$

It is well-defined because $\ker D_+ \subset \ker F^*$ was proved above. For $Cv = [u]$, its defining equation gives $\langle Cv, [\alpha] \rangle_+ = u^* D_+ \alpha = v^* F^* \alpha = \langle v, \Omega^{-1} F^* \alpha \rangle_\Omega$, proving the adjoint identity. Define $\mathcal{L} = 1_{\mathcal{C}_+} - CC^\dagger$. Then

$$\langle [\alpha], \mathcal{L}[\beta] \rangle_+ = \alpha^* (D_+ - F \Omega^{-1} F^*) \beta = \alpha^* (U_j - K_j) \beta, \quad (\text{RM.33})$$

$$\mathcal{L} \succeq 0, \quad J_+ \mathcal{L} J_+^\dagger = (U_j - K_j) G_i = Z_+ - Z. \quad (\text{RM.34})$$

The first equality follows by (RM.32) and the definition of the adjoint; the second follows by (RM.2) and (RM.20). The right-hand form is nonnegative by the actual upper-kernel comparison. Multiplying out the left side of (RM.34) gives $J_+ J_+^\dagger - J_+ C C^\dagger J_+^\dagger = Z_+ - F F^\dagger$, proving its last equality. Thus the complete excess of the comparator over the arithmetic cost is retained as a specific source contraction defect.

For clarity, the arithmetic upper kernel itself is formed on the original polynomial coordinates. Write $X_j = [[S^0]_\chi, \dots, [S^j]_\chi]$ and let H_j^m be the full monomial Gram for the actual source measure $m_{h,k}(u)du$, with $S = k/2 + iu$. The arithmetic upper-comparison proof constructs the explicit positive measure

$$d\nu_{h,k}(u) = a_k e^{-\alpha|u|} (k - 2 + |u|)^{-B} du, \quad \alpha = \pi/2, \quad B = 42 + 2 \deg h = 42 + 8m, \\ a_k = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)}, \quad k \geq 3,$$

where $m \geq 1$ is the full common multiplicity of the four zeros in the quartet and $\deg h = 4m$. Its literal constants are $c_h > 0$ and $\vartheta_h = \int_{-1}^1 w_h(t) dt > 0$. That complete proof establishes $d\nu_{h,k} \leq m_{h,k}(u)du$. For any polynomial coefficient vector v , $v^* H_j^\nu v = \int |\sum v_n (k/2 + iu)^n|^2 d\nu_{h,k} \leq v^* H_j^m v$. To prove the inverse inequality without a metric change, the exact completion of the square gives, for either positive form H ,

$$\ell^* H^{-1} \ell = \max_v \{2 \text{Re}(\ell^* v) - v^* H v\},$$

the maximum attained at $v = H^{-1} \ell$. The form inequality therefore implies $(H_j^m)^{-1} \preceq (H_j^\nu)^{-1}$. Consequently

$$U_j = X_j (H_j^\nu)^{-1} X_j^*, \quad K_j = X_j (H_j^m)^{-1} X_j^* \preceq U_j. \quad (\text{RM.35})$$

The expression for K_j agrees exactly with (RM.1): the invertible triangular monic-polynomial coefficient matrix transports H_j^m to O_j and X_j to B_j , so the middle matrices cancel in this composite. All factors $(k/2 + iu)^n$ remain in these Grams. The measure inequality and its constants are supplied by the complete arithmetic upper-comparison proof retained with this supplement; no bound on the actual omitted high-degree moments is assumed.

8 Attribution and exact source scope

The general trace-constrained determinant bound is classical. The author-hosted copy of Grone, Johnson, Marques de Sá and Wolkowicz, *Improving Hadamard's inequality*, Linear and Multilinear Algebra 16 (1984), 305–322, identifies the dimension/trace/trace-square optimization; see <https://www.math.uwaterloo.ca/~hwolkowi/henry/reports/PAPER17.pdf> and <https://doi.org/10.1080/03081088408817634>. The complete argument (RM.11)–(RM.19) above is independently supplied here. The added comparison consists of the exact maps (RM.20)–(RM.25).

The unchanged incoming archive has SHA256

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ec9324d794e839fac855eccea6a0c16e5e
736022441ac993a464df1f96b01d39
```

Its full note, markdown, handoff, checker, patch, source metadata and all 37 original file members remain preserved. The note's actual high-degree moment requirements remain stated; its unproved asymptotic example is recorded as motivation in the source, without supplying any missing arithmetic estimate. No Lean, actual zeta-packet quadrature, uniform trace-growth bound, or RH conclusion is asserted by this finite supplement.